radio units, in a data network.

- Each radio unit consists of a transceiver (a device that can both send and receive communication) and a highly directive antenna. The antenna only emits or receives power in the direction it is pointing.
- The two radio units are mounted and are directed towards each other with no obstacles, such as buildings, in between them that can hinder or disturb the transmission.
- As the connection is very directive it enables very high signal to noise ratio and thereby high data rates.
- The primary downside is that radio links require direct so called line-of-sight for optimum performance. Compared to fibre the connection is less stable as bad weather can interrupt the connection, in particular at higher frequencies.

Microwave Link

- Microwave is a line-of-sight wireless communication technology that uses high frequency beams of radio waves to provide high speed wireless connections that can send and receive voice, video and data information.
- One of the reasons microwave links are so adaptable is that they are broadband.
- They require no equipment or facilities between the two terminal points, so installing a microwave link is often faster and less costly than a cable connection.
- Microwaves are also able to penetrate rain, fog and snow, which means bad weather doesn't disrupt transmission.

Satellite link

- Satellite communication is one of the wireless technologies, used to transfer the signals from the transmitter to a receiver with the help of satellites.
- It is widely spread all over the world allowing users to stay connected virtually anywhere on the Earth.
- The Satellites used in this mode of communication, communicate directly with the orbiting satellites via radio signals.
- The process of satellite communication begins at an earth station. Here an installation is designed to transmit and receive signals from a satellite in orbit around the earth.
- The transmission system from the earth station to the satellite through a channel is called the **uplink**. The system from the satellite to the earth station through the channel is called the **downlink**.

Workshop-4

Identify the ways of pairing a mobile phone with other one using Bluetooth technology.

14.6. Network devices

Network devices are components used to connect computers or other electronic devices together so that they can share files or resources.